

PRE-CASE NEUROLOGICAL LOCALISATION SHEET

Anatomical structure → history clues → examination clues

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Use before solving cases: identify the anatomical level first. Ask what symptoms should be present if that structure is involved, then deliberately look for the matching positive examination signs — and for important findings that should be absent.

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Cerebrum & deep structures

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
Dominant frontal cortex — Broca area	Difficulty getting words out; short effortful speech; patient appears to know what they want to say; writing may also be impaired.	Nonfluent/effortful speech; impaired naming and repetition; comprehension relatively preserved; right lower facial/arm weakness may coexist.
Dominant posterior temporal cortex — Wernicke area	Fluent but meaningless speech; difficulty understanding conversation; family may report inappropriate answers.	Fluent paraphasic speech; poor comprehension; impaired repetition/naming; possible right superior quadrantanopia.
Dominant parietal cortex	Difficulty calculating/writing; right-left confusion; cannot recognise objects by touch; difficulty performing learned acts.	Acalculia, agraphia, finger agnosia, right-left disorientation; ideomotor apraxia; astereognosis/agraphesthesia.
Nondominant parietal cortex	Ignores one side; leaves food on one half of plate; bumps into objects on one side; poor spatial awareness.	Contralateral neglect; extinction on double simultaneous stimulation; constructional apraxia; anosognosia.
Medial frontal cortex / ACA territory	Leg-predominant weakness; reduced initiative/abulia; urinary urgency or incontinence.	Contralateral leg weakness > arm; hyperreflexia/Babinski; grasp reflex; frontal release signs.
Frontal eye field		

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
	Eyes noticed deviated to one side after acute event.	Conjugate gaze deviation toward cortical lesion and away from hemiparesis.
Supplementary motor area / medial frontal network	Difficulty initiating movement; limb may act independently; intermanual conflict.	Grasp reflex; alien-hand phenomena; impaired motor initiation; possible leg-predominant weakness.
Bilateral frontal opercula	Cannot voluntarily speak, swallow, cough or move lower face/tongue, but spontaneous emotional movements may persist.	Automatic-voluntary dissociation; severe voluntary faciopharyngoglossal weakness; preserved spontaneous smiling/yawning.
Occipital cortex	Loss of one side of visual world; bumping into objects; no ocular pain.	Contralateral homonymous hemianopia; high congruous defect; normal pupils/fundi; macula sparing may occur.
Bilateral occipital cortex	Behaves blind; may deny blindness or confabulate visual descriptions.	Cortical blindness with preserved pupillary reflexes and normal fundi; Anton syndrome (denial/confabulation).
Parieto-occipital association cortex — bilateral	Sees one object at a time; cannot take in a whole scene; misses objects when reaching.	Simultanagnosia + optic ataxia + ocular apraxia — Bálint syndrome.
Fusiform / inferior occipitotemporal cortex	Cannot recognise familiar faces but recognises people by voice, gait or clothing.	Prosopagnosia with preserved visual acuity and object perception.
Hippocampus / medial temporal lobes	Repeated questions; poor memory for recent events; remote memory relatively better.	

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
		Marked anterograde memory impairment; impaired delayed recall/new learning; attention and language may be preserved.
Bilateral anterior temporal lobes / amygdala	Hyperorality, hypersexuality, loss of fear, excessive visual exploration, behavioural change.	Klüver-Bucy pattern: placidity, hypermetamorphosis, visual agnosia, memory disturbance.
Internal capsule	Sudden dense face-arm-leg weakness with little/no cortical symptoms.	Pure motor hemiparesis; UMN face/arm/leg signs; hyperreflexia/Babinski; no aphasia/neglect/visual field defect.
Thalamus	Sudden hemisensory numbness; later burning pain/allodynia may appear.	Contralateral face-arm-leg sensory loss; pain relatively preserved; hyperpathia/allodynia; cheiro-oral pattern with small lesions.
Subthalamic nucleus	Sudden violent flinging movements of one side.	Contralateral hemiballismus; large-amplitude ballistic proximal movements.
Basal ganglia / substantia nigra	Slowness; rest tremor; reduced arm swing; softer voice; micrographia.	Bradykinesia, rigidity, rest tremor, masked facies, reduced arm swing, postural instability.
Temporal optic radiation — Meyer loop	Difficulty noticing objects in opposite upper visual quadrant.	Contralateral superior quadrantanopia — 'pie in the sky'.
Parietal optic radiation	Difficulty noticing objects in opposite lower visual quadrant.	Contralateral inferior quadrantanopia — 'pie in the floor'; parietal signs may coexist.

Cerebellum & ataxia

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
Cerebellar hemisphere	Clumsy same-side limb; spills drinks; overshoots targets; veers toward affected side.	Ipsilateral dysmetria, dysdiadochokinesia, intention tremor, rebound, heel-shin impairment.
Cerebellar vermis	Predominantly difficulty standing/walking; truncal swaying; little hand clumsiness.	Broad-based gait; truncal titubation; impaired tandem gait; limb coordination relatively preserved.
Sensory proprioceptive pathway	Unsteadiness worse in dark; needs to watch feet; stamping gait.	Loss of vibration/joint position; positive Romberg; pseudoathetosis; worsens on eye closure.

Visual pathway

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
Optic nerve	Monocular visual loss; colour desaturation; central blur; pain on eye movement if optic neuritis.	Reduced acuity; RAPD; impaired colour vision; central scotoma; disc may initially be normal.
Optic chiasm	Progressive peripheral visual loss; bumping into door frames; endocrine symptoms may coexist.	Bitemporal hemianopia; optic pallor later; ocular motility often initially normal.
Optic tract	Same visual hemifield lost in both eyes, often with deep cerebral symptoms.	Contralateral homonymous hemianopia; possible contralateral RAPD; associated thalamic/internal capsular signs.

Cranial nerves, brainstem & skull base

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
CN III — oculomotor nerve	Ptosis, diplopia, eye turned outward; ask about sudden severe headache.	Ptosis; eye down and out; impaired adduct elevation/depression; pupil involvement suggests compression.
CN IV — trochlear nerve	Vertical diplopia worse while reading or walking downstairs; compensatory head tilt.	Hypertropia; weak depression of adducted diplopia worsens with head tilt toward affected side.
CN VI — abducens nerve	Horizontal diplopia worse looking toward affected side.	Failure of ipsilateral abduction; isolated pal nerve/fascicle; both eyes fail to gaze toward side = VI nucleus/PPRF.
MLF	Horizontal diplopia; commonly demyelination in younger patients or stroke in older patients.	INO: impaired adduction on side of lesion - abducting nystagmus of opposite eye; convergence often preserved.
PPRF / abducens nucleus	Cannot look toward one side with either eye.	Ipsilateral conjugate horizontal gaze palsy; vertical gaze preserved.
Dorsal rostral midbrain	Difficulty looking upward; diplopia; headache/hydrocephalus symptoms may coexist.	Vertical gaze palsy; convergence-retraction nystagmus; light-near dissociation; lid retraction.

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
CN V — trigeminal	Facial numbness/pain; chewing difficulty.	V1/V2/V3 sensory loss; absent afferent corneal reflex; weak masseter/pterygoid; jaw deviation toward weak side.
CN VII — supranuclear	Sudden mouth deviation often with limb weakness.	Contralateral lower facial weakness with forehead sparing; eye closure preserved; accompanying UMN signs.
CN VII — peripheral / LMN	Whole side of face weak; cannot close eye; food/water leaks; ask taste, hyperacusis, lacrimation.	Ipsilateral upper + lower facial paralysis; proximal lesions may add taste loss, hyperacusis and reduced lacrimation.
CN VIII / vestibular pathway	Vertigo, oscillopsia, nausea; hearing loss/tinnitus suggests cochlear/AICA/CPA involvement.	Nystagmus; gait veering; head impulse abnormal in peripheral lesions; sensorineural hearing loss if cochlear component.
CN IX/X / nucleus ambiguus	Dysphagia, nasal regurgitation, choking, weak cough, hoarseness.	Palatal droop; uvula deviates away; reduced gag; dysphonia; weak cough; vocal-cord paralysis.
CN XI	Shoulder droop; difficulty shrugging or turning head; may follow neck surgery.	Trapezius wasting/weak shrug; SCM weakness causing weak head rotation to opposite side.
CN XII — LMN	Dysarthria; difficulty manipulating food in mouth.	Ipsilateral tongue wasting/fasciculations; tongue deviates toward lesion on protrusion.

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
Midbrain — medial/ventral	Ptosis/diplopia with opposite body weakness.	Ipsilateral III palsy + contralateral UMN weakness — Weber-type localisation.
Midbrain tegmentum	III-nerve symptoms plus opposite tremor/incoordination.	Ipsilateral III palsy + contralateral tremor/dysmetria/ataxia — red nucleus/superior cerebellar pathway.
Medial pons	Diplopia or facial weakness plus opposite limb weakness.	Ipsilateral VI and/or VII fascicular deficit + contralateral hemiparesis.
Dorsal pons / facial colliculus	Cannot look toward one side + same-sided complete facial weakness.	Ipsilateral conjugate gaze palsy + ipsilateral LMN VII palsy; ± contralateral weakness.
Dorsal pons — PPRF + MLF	Severe horizontal gaze difficulty.	One-and-a-half syndrome: gaze palsy toward lesion + INO when looking away.
PPRF/VI nucleus + MLF + VII fascicle	Horizontal gaze paralysis plus complete ipsilateral facial paralysis.	Eight-and-a-half syndrome = one-and-a-half syndrome + VII palsy.
Lateral pons / AICA territory	Vertigo, vomiting, ipsilateral hearing loss, facial weakness/numbness, gait unsteadiness.	Ipsilateral LMN facial palsy/deafness/facial pain-temperature loss/ataxia + contralateral body pain-temperature loss.
Lateral medulla	Vertigo, vomiting, dysphagia, hoarseness, hiccups, gait instability.	Ipsilateral facial pain-temperature loss + contralateral body pain-temperature loss; ipsilateral Horner/ataxia/palatal weakness.

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
Medial medulla	Opposite-sided weakness with tongue difficulty.	Ipsilateral LMN XII palsy + contralateral hemiparesis; contralateral proprioceptive loss may occur.
Bilateral ventral pons	Unable to move or speak but appears conscious.	Quadriplegia, anarthria, impaired horizontal gaze with preserved consciousness, blinking and vertical gaze — locked-in syndrome.
Cerebellopontine angle	Progressive unilateral hearing loss/tinnitus; later facial numbness and imbalance.	Ipsilateral sensorineural deafness; absent corneal reflex; facial weakness; ipsilateral ataxia/nystagmus.
Cavernous sinus	Painful ophthalmoplegia; retro-orbital headache; facial numbness.	Combined III/IV/VI palsies + V1 ± V2 loss; nerve generally spared; VI + Horner particularly localising.
Orbital apex	Painful visual loss plus diplopia/ptosis.	Optic neuropathy/RAPD + III/IV/VI and V1 abnormalities; visual loss distinguishes from cavernous sinus.
Jugular foramen	Hoarseness + dysphagia + shoulder weakness.	IX, X, XI deficits; palate/vocal cord + trapezius/SCM; XII spared — Vernet pattern.
Jugular foramen + hypoglossal region	Dysphagia, hoarseness, shoulder weakness and tongue weakness.	IX–X–XI–XII palsies — Collet-Sicard pattern.

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
Retroparotid/parapharyngeal space + sympathetic chain	Lower cranial-nerve symptoms plus ptosis/miosis.	IX–X–XI–XII palsies + Horner syndrome — Villaret pattern.
Extensive unilateral skull base	Slowly accumulating multiple cranial-nerve symptoms on one side with surprisingly normal limbs.	Multiple unilateral cranial neuropathies with long-tract signs — Garcin pattern.
Petrous apex	Chronic otitis/otorrhoea followed by deep facial/retro-orbital pain and diplopia.	Ipsilateral VI palsy + trigeminal V1/V2 pain sensory signs with ear infection — Graden pattern.
Intramedullary cord lesion	Poorly localised burning pain; gradual bilateral symptoms; bladder late.	Dissociated sensory loss; sacral sparing; segmental LMN signs at level + long-tract signs below.
Extramedullary cord compression	Local back pain → radicular pain → progressive leg weakness; sphincter symptoms late.	Early root signs; asymmetric long-tract abnormalities; Brown-Séquard pattern; sensory level develops.
Conus medullaris	Saddle numbness; very early urinary retention, bowel or sexual dysfunction; less radicular pain.	Symmetric saddle anesthesia; reduced anal tone; absent anal wink/bulbocavernosus; no UMN/LMN leg signs.

Roots, plexus, nerve, NMJ & motor neuron

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
Bilateral corticobulbar tracts	Dysarthria/dysphagia; inappropriate laughing/crying; multiple prior strokes may be present.	Spastic dysarthria; brisk jaw jerk; exaggerated gag; slow stiff tongue without wasting; emotional lability.
Cervical enlargement	Hand/arm weakness/wasting plus stiff legs; hand clumsiness and gait difficulty.	LMN signs in arms at lesion + UMN signs in legs below; sensory level; Hoffmann/Babinski may be present.
Anterior horn cell	Progressive painless focal wasting/weakness without numbness.	Segmental wasting, fasciculations, weakness and reduced reflexes; sensation normal.
Motor neuron system — UMN + LMN	Progressive weakness spreading between regions; cramps/fasciculations; bulbar symptoms; no sensory complaints.	Wasting/fasciculations + brisk reflexes/Babinski; tongue LMN signs possible; sensation and sphincters usually preserved.
Cervical nerve root	Neck pain radiating down arm; cough/sneeze may worsen; paresthesia in a dermatome.	Myotomal weakness; reduced corresponding reflex; dermatomal sensory loss; Spurling's test to reproduce symptoms.
C6 root	Pain/paresthesia lateral arm to thumb.	Weak elbow flexion/wrist extension; reduced biceps/brachioradialis reflex; thumb/lateral forearm sensory loss.

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
C7 root	Posterior arm pain to middle finger.	Weak triceps/finger extension/wrist flexion; reduced triceps reflex; middle-finger sensory loss.
C8 root	Medial forearm pain to ring/little fingers.	Weak finger flexion/intrinsic hand muscles across median + ulnar nerves; C8 sensory pattern.
L5 root	Back pain radiating down lateral leg to dorsum of foot; foot drop.	Weak dorsiflexion plus inversion and hip abduction; L5 sensory loss; ankle jerk usually preserved.
S1 root	Back pain radiating down posterior leg/lateral foot; difficulty standing on toes.	Weak plantarflexion; reduced/absent ankle reflex; lateral-foot sensory loss; difficulty toe walking.
Lower trunk brachial plexus C8–T1	Shoulder/arm pain followed by hand weakness; medial forearm/hand numbness.	Weakness crossing median + ulnar territories; medial forearm sensory loss; Horner may coexist in Pancoast lesion.
Common peroneal nerve	Foot drop after leg crossing, squatting or weight loss; usually no back pain.	Weak dorsiflexion + eversion; inversion preserved; sensory loss anterolateral leg/dorsum foot; ankle jerk normal.
Sciatic nerve	Hip/buttock trauma or surgery followed by posterior-leg symptoms and foot weakness.	Dorsiflexion and plantarflexion weak; inversion/eversion weak; ankle jerk reduced; hamstring involvement may occur.

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
Radial nerve — spiral groove	Wrist drop after prolonged arm compression.	Wrist/finger extension weakness; triceps preserved; small first dorsal web-space sensory deficit.
Posterior interosseous nerve	Finger drop without numbness.	Finger/thumb extension weak; wrist extensors preserved but may deviate radially; no sensory loss.
Anterior interosseous nerve	Difficulty pinching/picking up small objects; no numbness.	Cannot make normal 'OK' sign; weak FPL, index FDP; pure motor lesion.
Neuromuscular junction — postsynaptic	Fluctuating ptosis/diplopia; chewing/speech worsens with use; better after rest.	Fatigable weakness; variable ophthalmoplegia; pupils normal; sensation normal; reflexes preserved.
Neuromuscular junction — presynaptic	Proximal weakness improves after activity; dry mouth and autonomic symptoms; smoking history relevant.	Reduced reflexes that augment after exercise; proximal weakness; autonomic signs.
Diffuse peripheral roots/nerves	Ascending weakness after infection; paresthesias; facial weakness; autonomic symptoms.	Symmetric flaccid weakness + generalized areflexia; mild sensory deficit; facial diplegia; autonomic instability.

Spinal cord, conus & cauda

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
High cervical cord C1–C4	Neck pain; hand/leg weakness progressing to quadriparesis; breathing difficulty.	UMN weakness in all four limbs; sensory level at C2–3; diaphragmatic weakness at C3–5; cranial nerves normal.
C5–C6 cervical cord	Neck/arm symptoms with progressive hand clumsiness and leg stiffness.	Reduced biceps/brachioradialis reflex at elbow but brisk triceps/fingers/legs below; inverted supinator may occur.
Thoracic cord	Progressive leg heaviness/stiffness; band-like truncal sensation; urinary urgency; arms unaffected.	Spastic paraparesis; brisk knees/ankles; bilateral Babinski; truncal sensory level; arms normal.
Dorsal columns	Unsteadiness worse in dark; needs visual guidance; electric sensations possible.	Loss of vibration/proprioception; positive Romberg; stamping gait; pseudoathetosis.
Spinothalamic tract	Poor appreciation of pain/temperature; unnoticed burns/injuries.	Contralateral pain-temperature loss below lesion, often starting a few segments below
Anterior cord	Sudden severe weakness, often vascular/aortic setting; bladder involvement.	Bilateral paralysis/UMN signs + pain-temperature loss; vibration and position preserved.

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
Posterior cord	Severe sensory ataxia with relatively preserved strength and pain sensation.	Loss of vibration/proprioception; positive Romberg; pain-temperature comparatively preserved.
Hemicord	Trauma/compression with unilateral weakness and sensory symptoms.	Ipsilateral UMN weakness + ipsilateral vibration/proprioception loss + contralateral pain-temperature loss — Brown-Séquard.
Central cervical cord / syrinx	Burns hands unknowingly; progressive hand weakness; cape-like numbness.	Bilateral segmental pain-temperature loss + preserved vibration; hand wasting/areflexia later UMN legs.
Acute transverse cord lesion	Rapid bilateral weakness; numbness rising to a level; urinary retention.	Bilateral motor deficit + clear sensory level + autonomic dysfunction; initially flaccid, later spastic.
Spinal shock	Immediately after severe acute spinal injury: complete paralysis below lesion.	Flaccidity and areflexia below lesion despite cord disease; bulbocavernosus initially absent later hyperreflexia/spasticity.
Conus medullaris	Distal leg weakness prominent; sphincter symptoms less abrupt than conus.	Mixed LMN distal legs + possible Babinski/UMN findings; ankle jerks often absent; bladder less dominant initially.
Cauda equina		

Anatomical structure / localisation	Positive history findings / what to ask	Positive examination findings / what to elicit
	Severe back/radicular leg pain; asymmetric weakness/numbness; bladder dysfunction often later.	Asymmetric LMN weakness; reduced/absent reflexes; flexor plantars; patchy root sensory loss; saddle anesthesia.

Educational bedside localisation aid · 3-column pre-case reference sheet